

MTH313M Quiz Part B (April 7, 2026)

6:45 pm - 7:45 pm (+20 minutes for DAP students)

Instructions:

1. Write your answers in the designated place. If your answer is not legible, you shall not get credit.
2. DO NOT attach any additional sheet to this page.
3. Read the question stem and answer the sub-questions separately in the space provided.
4. Provide all details/justification including calculations.

Name:

Roll No.:

Question 3. Suppose that earthquakes in a given season at a particular place occur following a Poisson Process with parameter $\lambda_1 > 0$ or $\lambda_2 > 0$. It occurs with parameter λ_1 for $100p\%$ of the time and with parameter λ_2 for $100(1 - p)\%$ of the time, where $p \in (0, 1)$ is known.

- (a) (3 marks) Describe the distribution of the number of earthquakes during the first t units of time, $t > 0$.
- (b) (2 marks) Given that 4 earthquakes occurred during the first 10 units of time, find the conditional probability that the parameter value was λ_2 .

Question 4. Let $\{N_t\}_{t \geq 0}$ be a Poisson Process with rate $\lambda > 0$. Fix $0 < r < s$ and let n be a positive integer.

- (a) (3 marks) Find the conditional distribution of N_r given $N_s = n$.
- (b) (1 + 1 marks) Compute the mean and variance of this conditional distribution.